



ES 19 PROBLEM 1 PREPARATION NOTES

ATTENTION

CONSTRAIN DISPLACEMENT

2 DOF

{ DON'T REPLACE F WITH a FORCE! }



you may have constrain displacement as input applied to spring/damper constrain
 ↳ SHORTCUT → WORKS ONLY IF you remove and replace as a force applied on the hinge

applied on spring + damper

$$y = y_0 e^{i\Omega t}$$

$$\dot{y} = i\Omega y_0 e^{i\Omega t}$$

HARMONIC displacement

$$(k_2 + i\Omega c_2) y_0 e^{i\Omega t} = F_y$$

new forcing
 complex magnitude force

F applied directly on → CONSTRAIN act on body ⇒ NO SHORTCUT simplification
 NO RIGID body! ⇒ we can only APPROACH it by extending coordinate vector
 (NOT on a spring)

2 indep coord θ, φ + constrain displacem.

$$\underline{x} = \begin{bmatrix} \theta \\ \varphi \\ y \end{bmatrix}$$

hinge subjected to constrain displacement

you define $[\lambda], \dots$

with 3 COLUMNS

3x3 Matrix

→ PARTITIONING of the

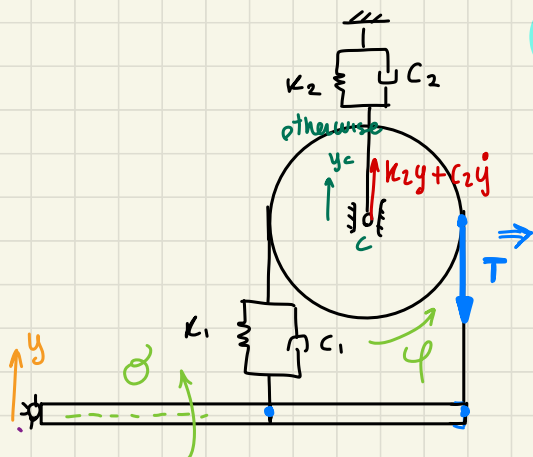
$$\underline{x} = \begin{bmatrix} \underline{x}_F \\ \underline{x}_c \end{bmatrix}$$

$$\begin{bmatrix} M_{FF} & M_{Fc} \\ M_{cF} & M_{cc} \end{bmatrix}$$

you cannot simplify the problem!

! due to excitation of constrain

• OUTPUT constraint forces



T?

OUTPUT FRF
respect TENSION
on the WIRE

constraint forces output



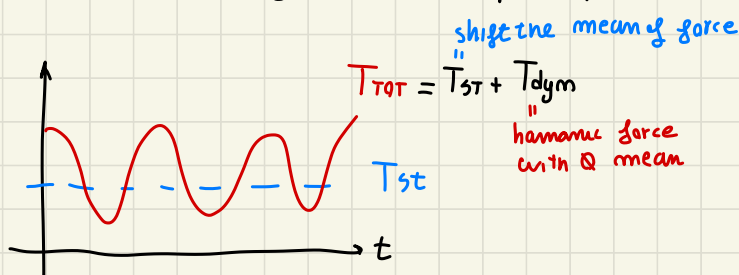
$$T = T_{ST} + T_{dym}$$

static component
Related to
• weights
• static elastic forces
↓
constant values

variable in
time, depending
on excitation
applied

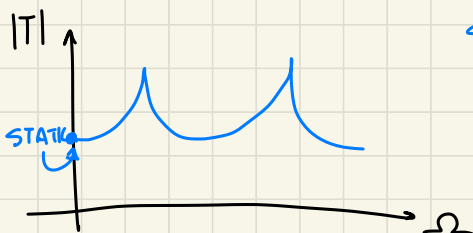
T_{dym} : time varying

IF OUTPUT is **time history**: we can superimpose $T_{ST} + T_{dym}$



IF OUTPUT is **FRF** → it is an I/O Linear Relationship (AFTIO)
relation I/O @ given frequency range

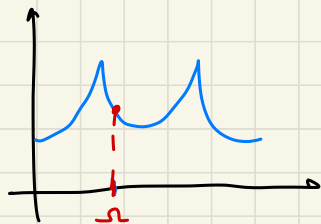
↓ you cannot MIX Different components dynamic!



static component
is the ω value

while the other values
are just for

$1/\Delta$ of dym components ω



IF mixing
 $T_{ST} + T_{dym}$
 @ $\Omega \rightarrow$ **WRONG!**

the offset is
 added $\forall \Omega$

$\Downarrow$ STATIC, **DYNAMIC**

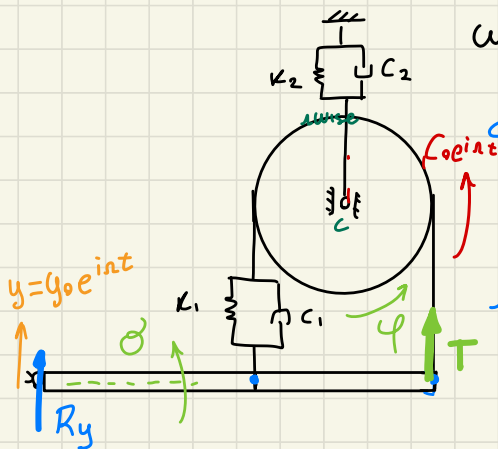


IT'S a
 transformation...
WRONG!

I just take
 dynamic
 part!

FRF

• When computing **CONSTRAIN FORCES**



We solve as $\underline{x} = \begin{pmatrix} \theta \\ \varphi \\ y \end{pmatrix}$

compute R_y

FRF or TIME HISTORY

$\Downarrow$
 you have 2 choices

1) using the **PARTITIONING**
 to obtain R_y

$M_{CF}, K_{CF}, C_{CF}, M_{FF}, K_{FF}, C_{FF}$

Where the force Reaction is due to
 y

IF $y_0 = 0$
 and we have Coe^{int}

best way is to use **D'ALAMBERT** on the system

FOR EX T?



$x_{G1} \approx 0$
 around
 equilibrium

easily computed as

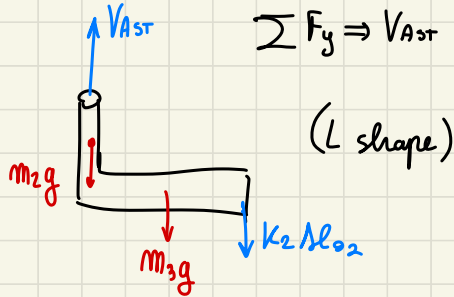
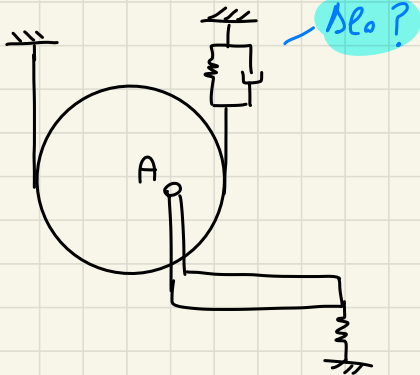
$$\sum_{BAR} M_0 = 0$$

$T = \dots$ expressing $y_{01}, \Delta \ell_1 \rightarrow T_{ST} + T_{dyn}$

IF cutting constrain between bodies

$\hookrightarrow$ CONSTRAIN FORCES! you miss pieces if you forget it

for EXAMPLE on MIDTERM



$$\sum F_y \Rightarrow V_{AST}$$

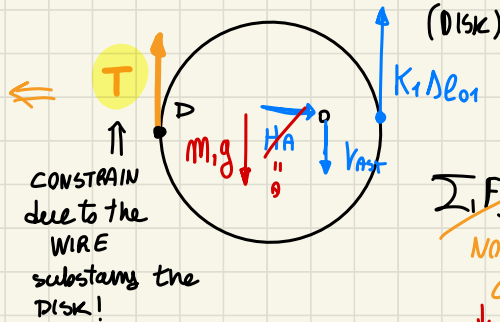
(L shape)

$$\sum_i M_D = Q$$

DISK

is the good chance that neglect the T force unknown!

only $\Delta \ell_0$ unknown



$$\sum_i F_y = 0$$

NOT good approach

$\downarrow$ 2 unknowns $T, \Delta \ell_0 \dots$

